Packages and access modifiers in java

- Package in java can be categorized in two form, built-in package and user-defined package.
- There are many built-in packages such as java, lang, awt, javax, swing, net, io, util, sql etc.

Advantage of Java Package

1) Java package is used to categorize the classes and interfaces so that they can be easily maintained.

2) Java package provides access protection.

3) Java package removes naming collision.

The **access modifiers** in java specifies accessibility (scope) of a data member, method, constructor or class.

There are 4 types of java access modifiers:

- ✓**private**
- ✓**default**
- ✓**protected**
- ✓**public**

1) private access modifier

The private access modifier is accessible only within class.
Simple example of private access modifier

```
class A{
private int data=40;
private void msg(){System.out.println("Hello java");}
}

public class Simple{
 public static void main(String args[]){
   A obj=new A();
   System.out.println(obj.data);//Compile Time Error
   obj.msg();//Compile Time Error
   }
}
```

Role of Private Constructor

If you make any class constructor private, you cannot create the instance of that class from outside the class.

```
class A{
private A(){}//private constructor
void msg(){System.out.println("Hello java");}
}
public class Simple{
 public static void main(String args[]){
   A obj=new A();//Compile Time Error
 }
}
```

3) protected access modifier

The **protected access modifier** is accessible within package and outside the package but through inheritance only.

The protected access modifier can be applied on the data member, method and constructor. It can't be applied on the class.

4) public access modifier

The **public access modifier** is accessible everywhere. It has the widest scope among all other modifiers.

Access Modifier	within class	within package	outside package by subclass only	outside package
Private	Y	N	N	N
Default	Y	Y	N	N
Protected	Y	Y	Y	N
Public	Y	Y	Y	Y

Access protection in packages:

Access protection	Description
No modifier (default)	The classes , methods and member variables specified in the same package are accessible to all the classes inside the same package.
public	The classes, methods and member variables under this modifier can be accessed from anywhere.
protected	The classes, methods and member variables under this modifier are accessible by all subclasses, and accessible by code in same package.
private	The methods and member variables are accessible only inside the class.

Java access modifiers with method overriding

If you are overriding any method, overridden method (i.e. declared in subclass) must not be more restrictive.

```
class A{
protected void msg(){
System.out.println("Hello java");}
}

public class Simple extends A{
void msg(){System.out.println("Hello java");}//C.T.Error

 public static void main(String args[]){
   Simple obj=new Simple();
   obj.msg();
   }
}
```